Spreadsheet verification of the library

Generated by `python3 tools/research/spreadsheet_check.py` (research log D94): for every project with a verification workbook, KAPSAM writes the workbook, LibreOffice Calc recalculates it and the rows of its sheet Check are compared with their tolerances.

189 workbooks, 4678 compared values, of which 3684 are computed by spreadsheet formulas and 994 repeat an input or a fixed factor (a check of the transcription, not of the arithmetic).

Project	values (computed)	largest difference / tolerance	worst quantity	result
bettencourt2026-fluoranthene-arithmetic	7 (5)	0	u (s_I⟨I⟩)	OK
bettencourt2026-fluoranthene-weighted	7 (5)	0	u (s_I⟨I⟩)	OK
bipm-isrd-01-qnmr-dmso2-ma-d2o	28 (12)	0.00019	c(N_A)	OK
bipm-isrd-01-qnmr-dmtp-ma-dmso-d6	28 (12)	0.00026	c(M_S)	OK
bipm-isrd-02-qnmr-dmtp-dmso2-cdcl3	28 (12)	0.00018	c(m_S)	OK
bipm-isrd-02-qnmr-ma-dmso2-d2o	28 (12)	0.00022	c(R)	OK
bipm-isrd-06-qnmr-ba-btmsb-d4-cd3od	28 (12)	0.00021	c(R)	OK
bipm-isrd-06-qnmr-dmso2-btmsb-d4-dmso-d6	28 (12)	8e-05	c(R)	OK
bipm-isrd-07-qnmr-dmso2-dss-d6-d2o	28 (12)	7.9e-05	c(N_A)	OK
bipm-isrd-07-qnmr-ma-dss-d6-d2o	28 (12)	0.00013	c(m_A)	OK
bipm-isrd-08-qnmr-btmsb-d4-ba-cd3od	28 (12)	0.00027	c(N_S)	OK
emue-e3.4-soil-lead-iso21748-rms-bias	21 (14)	1.3e-06	c(d_bias)	OK
emue-e3.4-soil-lead-iso21748	21 (14)	1.4e-06	u(y)	OK
emue-e3.8-ammonia-permeation	49 (29)	0.00014	c(x2_CO)	OK
emue-e6.5-ephedrine-wada-topdown	7 (6)	0	u (u(Rw), mean of 3)	OK
gum5-e1-ph-temperature-interpolation-correlated	32 (14)	0.00013	c(E_S2)	OK
gum5-e1-ph-temperature-interpolation	32 (14)	0.00016	c(E_S2)	OK
gum5-e1-ph-two-point-25c-correlated-standards	19 (9)	3.2e-05	c(E_S2)	OK
gum5-e1-ph-two-point-25c	19 (9)	4e-05	c(E_S2)	OK
gum5-e12-between-bottle-homogeneity-bayes	7 (7)	1.5e-05	s_x (B.7)	OK
gum5-e2-benzo-a-pyrene	14 (6)	0.00079	c(AISE)	OK
gum5-e3-molar-mass-glucose	13 (13)	0.0013	c(A_C)	OK

Project	values (computed)	largest difference / tolerance	worst quantity	result
gum5-e5-ghg-inventory-agriculture-shared-activity	34 (14)	4.8e-07	c(A5)	OK
gum5-e5-ghg-inventory-agriculture	40 (16)	4.1e-07	c(F5)	OK
iso13528-e1-censored-deleted	49 (49)	6.9e-07	z (L)	OK
iso13528-e1-censored-half	59 (59)	0	median	OK
iso13528-e1-censored-ignored	59 (59)	5.3e-07	z (J)	OK
iso13528-e2-arsenic-homogeneity	11 (11)	2.8e-05	$\lvert \bar{y}_1 - \bar{y}_2 \rvert$	OK
iso13528-e3-atrazine-algorithm-a	81 (81)	2.9e-06	z (6)	OK
iso13528-e3-atrazine-median	81 (81)	0	median	OK
iso13528-e4-mercury-feed	97 (97)	0	median	OK
iso13528-e5-los-angeles-reference-value	11 (9)	2.3e-06	c(x_CRM)	OK
iso21748-c2-meat-content	20 (10)	0.00013	c(f_N)	OK
iso21748-c4-crude-fibre	8 (8)	0	u (drying to constant mass)	OK
iso5725-2-c1-sulfur-in-coal-reml	76 (76)	2.3e-05	k (4, 2)	OK
iso5725-2-c1-sulfur-in-coal	92 (92)	8.2e-06	h (4, 2)	OK
iso5725-2-c3-creosote-oil-reml	105 (105)	0	h (1, 1)	OK
iso5725-2-c3-creosote-oil	125 (125)	0	h (1, 1)	OK
iso5725-3-j1-carbon-steel	4 (4)	0	u (s_I(TO))	OK
iso5725-3-j2-vanadium-steel	3 (3)	0	k	OK
iupac2024-4a-water-hardness	16 (9)	5.7e-07	c(g_Ca)	OK
iupac2024-4h-cadmium-standard-beta-mcm	13 (9)	0.00011	u(y)	OK
iupac2024-4i-argentometric-titration	19 (14)	3.2e-05	c(dV_E)	OK
iupac2024-5c-benzoic-acid-purity-beta	7 (7)	0.0016	U	OK
mupr-b1-nitrate	9 (7)	0	u (s_I⟨I⟩)	OK
mupr-b2-pooled-precision	7 (7)	0	u (s_I(P)⟨I⟩)	OK
mupr-b2-replicates-on-two-days	6 (6)	0	u (s′_I(P)⟨II⟩)	OK
mupr-b3-linear-precision	5 (5)	0	u (u′(DSS))	OK
mupr-b4-arsenic-one-crm	5 (5)	0	u (s_I⟨I⟩)	OK
mupr-b5-crm-and-spiked-samples	4 (4)	0	u (s′_I⟨II⟩)	OK
mupr-b6-sediment-22-pt	9 (9)	0	u (s_I(DS)⟨I⟩)	OK
mupr-ex2-reference-materials	3 (3)	0	k	OK

Project	values (computed)	largest difference / tolerance	worst quantity	result
mupr-ex3-spiked-samples	3 (3)	0	k	OK
nist-tn1900-e8-molecular-weight-co2	10 (10)	0.0027	u(y)	OK
nist-tn1900-e9-cadmium-standard	13 (11)	5.1e-05	c(P)	OK
nordtest-3-ammonium-pt	7 (7)	0	u (u(Rw))	OK
nordtest-5.3-ammonium-duplicates	9 (7)	0	u (u(Rw), control < 30)	OK
nordtest-5.3-oxygen-duplicates	5 (4)	0	u (s_r, duplicates)	OK
nordtest-6.1-one-crm	6 (6)	0	RMS_bias (u(bias))	OK
nordtest-6.1-three-crms	6 (6)	0	RMS_bias (u(bias))	OK
nordtest-6.2-proficiency-tests	6 (6)	0	RMS_bias (u(bias))	OK
nordtest-6.3-recovery	6 (6)	0	RMS_bias (u(bias))	OK
nordtest-8-bod-crm	7 (7)	0	u (u(Rw))	OK
nordtest-8-bod-pt	7 (7)	0	u (u(Rw))	OK
nordtest-8-pcb-crm	7 (6)	0	u (u(Rw))	OK
nordtest-8-pcb-pt	7 (6)	0	u (u(Rw))	OK
palma2019-cr-marine-sediment	13 (9)	0	u (s_r(h)(l))	OK
palma2019-cu-marine-sediment	13 (9)	0	u (s_r(h)(l))	OK
palma2019-li-marine-sediment	13 (9)	0	u (s_r(h)(l))	OK
palma2019-mn-marine-sediment	13 (9)	0	u (s_r(h)(l))	OK
palma2019-ni-marine-sediment	13 (9)	0	u (s_r(h)(l))	OK
palma2019-pb-marine-sediment	13 (9)	0	u (s_r(h)(l))	OK
palma2019-zn-marine-sediment	13 (9)	0	u (s_r(h)(l))	OK
possolo-bodnar-2018-chromium-soil-homogeneity-bayes	7 (7)	1.4e-08	F_m	OK
possolo-meija-2022-plasma-anion-gap	16 (8)	7.2e-06	c(c_Na)	OK
quam-a1-cadmium-standard-volume	21 (16)	5.9e-05	c(dV_cal)	OK
quam-a1-cadmium-standard	13 (9)	0.00011	c(V)	OK
quam-a2-naoh-standardisation-submodels	34 (30)	0.00033	c(m_gross)	OK
quam-a2-naoh-standardisation	19 (9)	1.8e-05	c(rep)	OK
quam-a3-hcl-titration	25 (11)	5.7e-05	c(P_KHP)	OK
quam-a4-pesticides-combination	13 (7)	0.00032	c(Rec)	OK
quam-a4-pesticides-topdown	5 (5)	0	u (precision)	OK

Project	values (computed)	largest difference / tolerance	worst quantity	result
quam-a5-cadmium-release	22 (10)	0.00037	c(a_V)	OK
quam-a6-crude-fibre	11 (8)	0	u (s_R (2.5))	OK
quam-a7-assay-standard	22 (10)	0.00035	c(m2)	OK
quam-a7-lead-double-idms	103 (37)	6.4e-05	c(R_x1)	OK
ufs-a6-cadmium-soil	29 (19)	1.3e-06	c(f_dry)	OK
ufs-a6-phosphorus-soil	26 (18)	6.1e-07	c(f_split)	OK
ume-procedure-19-2-qnmr-purity-internal-standard	30 (21)	6e-05	c(m_std)	OK
ea402-s3-resistor-10kohm	23 (22)	0.15	c(r_C)	OK
ea402-s6-power-sensor-18ghz	32 (27)	5.9e-05	c(M_Sc)	OK
ea402-s7-step-attenuator-30db	32 (25)	6.7e-05	c(dL_0a)	OK
ea402-s9-dmm-100v	15 (13)	8.5e-05	c(V_S)	OK
gum-h2-impedance	13 (7)	8.2e-05	u(y)	OK
gum-h2-reactance	13 (7)	3.6e-05	c(V)	OK
gum-h2-resistance	13 (7)	0.00024	U	OK
jcgm101-9.4-comparison-loss-x000-correlated-higher-order	8 (4)	0	x(X1)	OK
jcgm101-9.4-comparison-loss-x000-correlated	8 (4)	0	x(X1)	OK
jcgm101-9.4-comparison-loss-x000	8 (4)	0	x(X1)	OK
jcgm101-9.4-comparison-loss-x010-correlated-higher-order	8 (4)	5.7e-08	c(X1)	OK
jcgm101-9.4-comparison-loss-x010-correlated	10 (6)	5.8e-08	U	OK
jcgm101-9.4-comparison-loss-x010	8 (4)	5.7e-08	c(X1)	OK
jcgm101-9.4-comparison-loss-x050-correlated-higher-order	8 (4)	4.5e-07	c(X1)	OK
jcgm101-9.4-comparison-loss-x050-correlated	10 (6)	4.6e-07	U	OK
jcgm101-9.4-comparison-loss-x050	8 (4)	4.5e-07	c(X1)	OK
m3003-c5-dvm-dominant-rectangular	11 (10)	4.6e-05	u(y)	OK
m3003-k1-resistor-10kohm	25 (23)	0.44	ν_eff	OK
m3003-k2-power-sensor-18ghz	32 (30)	6.2e-06	ν_eff	OK
m3003-k3-attenuator-30db-mismatch-e5	26 (23)	8.7e-05	c(dR_D)	OK
m3003-k3-attenuator-30db	26 (23)	7.2e-05	c(dR_D)	OK
m3003-l3-dmm-1v-range-095v	31 (27)	0	x(I_DMM)	OK
m3003-l3-dmm-1v-range-1v-dominant	29 (25)	0	x(I_DMM)	OK

Project	values (computed)	largest difference / tolerance	worst quantity	result
m3003-l3-dmm-1v-range-500mv	31 (27)	0	x(I_DMM)	OK
num-10-resistance	13 (7)	0.00034	c(phi)	OK
possolo-meija-2022-wheatstone-bridge	16 (8)	3e-05	c(R_E)	OK
ume-procedure-19-1-dmm-10v-calibration	17 (14)	1.9e-07	ν_eff	OK
cg19-ex1-flask-1000ml	38 (30)	0.0073	c(d_res)	OK
cg19-ex2-pycnometer-100ml	37 (28)	0.0065	c(d_form)	OK
ea402-s12-water-meter-error	15 (12)	0.00013	u(y)	OK
ea402-s12-water-meter-mean-error	11 (9)	6.9e-08	U	OK
ea402-s12-water-meter-volume	28 (27)	0.00016	c(dV_iS)	OK
m3003-k9-flowrate-standard-conditions	58 (49)	1e-05	c(i)	OK
nist-tn1900-e4-pitot-tube	15 (8)	6.6e-05	c(p)	OK
gum5-e9-torque-slope-ols	29 (13)	6e-05	c(Y4)	OK
gum5-e9-torque-slope-wls	29 (13)	0.00011	c(Y3)	OK
iso21748-c1-co-emissions	3 (3)	0	k	OK
iso21748-c3-aerobic-plate-count-flour	4 (4)	0	u (sample preparation)	OK
iso21748-c3-aerobic-plate-count-shrimp	4 (4)	0	u (sample preparation)	OK
iso21748-c3-aerobic-plate-count-vegetables	4 (4)	0	u (sample preparation)	OK
jcgm101-9.2.2-additive-gaussian	16 (8)	0	x(X1)	OK
jcgm101-9.2.3-additive-rectangular	16 (16)	0	x(X1)	OK
jcgm101-9.2.4-additive-dominant-rectangular	16 (16)	0	x(X1)	OK
num-3-ratio-abc	13 (11)	0.089	c(c)	OK
ea402-s10-vernier-calliper	18 (17)	0.00032	c(l_S)	OK
ea402-s13-ring-gauge-90mm	29 (22)	5.2e-08	c(dl)	OK
ea402-s13-ring-gauge-temperature	23 (23)	0	x(a_S)	OK
ea402-s4-gauge-block-50mm	28 (23)	0.11	c(dl_C)	OK
gum-h1-end-gauge	26 (20)	0.071	c(d_theta)	OK
jcgm101-9.5-gauge-block-approximate	32 (22)	0.085	c(D)	OK
jcgm101-9.5-gauge-block-nonlinear	32 (22)	0.11	c(d_theta)	OK
m3003-k10-sheet-thickness	25 (23)	1e-05	u(y)	OK
m3003-k6-gauge-block-10mm-temperature-model	33 (29)	0.02	c(dC)	OK

Project	values (computed)	largest difference / tolerance	worst quantity	result
m3003-k6-gauge-block-10mm	26 (24)	0.02	c(dC)	OK
num-8-end-gauge-gaussian	22 (10)	0.12	c(d_theta)	OK
possolo-meija-2022-storage-tank-volume	10 (6)	1.5e-05	c(R)	OK
cg18-h1-100g-option2	40 (29)	0.038	c(m_c)	OK
cg18-h1-220g-max-option1-delta-t	28 (22)	0.043	ν_eff	OK
cg18-h1-220g-max-option1	28 (24)	0.037	ν_eff	OK
cg18-h1-220g-max-option2	40 (29)	0.059	c(dI_ecc)	OK
cg18-h1-220g-zero	13 (11)	0	x(I_L)	OK
cg18-h2-60kg-10kg-option1	30 (27)	0.00043	c(dm_D)	OK
cg18-h2-60kg-max-adjusted	30 (27)	0.00057	c(dm_D)	OK
cg18-h2-60kg-max-option1-delta-t	30 (25)	0.00059	c(dI_rep)	OK
cg18-h2-60kg-max-option1	30 (27)	0.00059	c(dI_rep)	OK
cg18-h2-60kg-max-option2	37 (28)	0.00059	c(dI_rep)	OK
cg18-h2-60kg-zero	13 (11)	0	x(I_L)	OK
cg18-h3-30t-10t-weights	31 (29)	0.00012	c(dm_B)	OK
cg18-h3-30t-15t-substitution	64 (60)	8.1e-05	c(dm_c)	OK
cg18-h3-30t-max-substitution	96 (89)	9.9e-05	c(dm_c)	OK
cg18-h3-30t-zero	13 (11)	0	x(I_L)	OK
ea402-s2-weight-10kg	19 (16)	7.5e-08	c(dm_D)	OK
ilac-p14-cmc-floor-weight-10kg	19 (16)	7.5e-08	c(dm_D)	OK
jcgm101-9.3-mass-calibration-higher-order	17 (13)	0.044	c(m_Rc)	OK
jcgm101-9.3-mass-calibration	19 (15)	0.044	c(m_Rc)	OK
m3003-k4-weight-10kg	23 (21)	0.024	c(dA_b)	OK
m3003-k5-weighing-machine-200g	25 (22)	0.016	c(W_S)	OK
nist-tn1900-e1-weighing	13 (7)	0.00013	U	OK
gum5-e14-radioactive-decay-bi207	12 (7)	0.00084	c(T_half)	OK
gum5-e15-wire-rope-breaking-force	20 (20)	0.00045	ν_eff	OK
iso5725-2-c2-softening-point-pitch-reml	137 (137)	2e-05	Grubbs, highest (2)	OK
iso5725-2-c2-softening-point-pitch	153 (153)	2e-05	Grubbs, highest (2)	OK
nist-tn1900-e13-thermal-expansion-gaussian-95	16 (8)	0.00047	c(L_1)	OK

Project	values (computed)	largest difference / tolerance	worst quantity	result
nist-tn1900-e13-thermal-expansion-gaussian-99	16 (8)	0.00047	c(L_1)	OK
nist-tn1900-e3-falling-ball-viscometer	22 (10)	0.019	c(t_C)	OK
num-11-stefan-boltzmann	19 (9)	0.087	c(R_inf)	OK
emue-e2.2-vacuum-gauge-correction-applied	13 (9)	4.3e-05	u(y)	OK
emue-e2.2-vacuum-gauge-correction-ignored	10 (8)	1.1e-05	c(p_UUC)	OK
emue-e2.2-vacuum-gauge-correction-in-uncertainty	13 (9)	2.4e-05	c(dp_m)	OK
emue-e2.2-vacuum-gauge-sum-model-correlated	22 (12)	2.4e-05	u(y)	OK
emue-e2.2-vacuum-gauge-sum-model-correlation-ignored	25 (13)	2.7e-05	u(y)	OK
m3003-k8-pressure-indicator-2mpa	50 (42)	4.4e-08	ν_eff	OK
ea402-s11-block-calibrator-180c	28 (25)	8.5e-05	c(t_S)	OK
ea402-s5-thermocouple-emf	25 (19)	0.00014	c(V_iX1)	OK
ea402-s5-thermocouple-furnace	28 (24)	8.8e-05	c(t_0S)	OK
m3003-k11-gas-temperature	26 (24)	3.2e-05	c(I_u)	OK
m3003-k11-probe-calibration	26 (24)	4.2e-05	u(y)	OK
m3003-k7-thermocouple-emf	60 (50)	0.00016	c(dt_0S)	OK
m3003-k7-thermocouple-furnace	33 (28)	8.8e-05	c(dt_0S)	OK
nist-tn1900-e2-surface-temperature	8 (8)	1.3e-06	c(t)	OK

Workbooks outside a tolerance: 0.